

DM — Scene 2 — Diabetes Pharmacology: The Gut-Kidney Plumbing



1. GLP-1 / GIP incretin jar

WHAT YOU SEE

Monk holding GLP-1 GIP jar in the gut tube, WEIGHT NEUTRAL/loss

WHAT IT ENCODES

GLP-1 receptor agonists (the '-tides': liraglutide, semaglutide, dulaglutide, exenatide) and the dual GIP/GLP-1 agonist tirzepatide augment glucose-dependent insulin secretion, suppress glucagon, slow gastric emptying, and increase satiety. The result is potent A1c lowering (~1-2%) with marked weight LOSS and low intrinsic hypoglycemia risk. GLP-1 RAs with CV-outcome data (semaglutide, dulaglutide, liraglutide) also reduce atherosclerotic MACE.

BOARD TRAP

Wrong answer: 'GLP-1 agonists cause weight gain like sulfonylureas and insulin.' Discriminator — GLP-1 RAs/tirzepatide produce weight loss (a key class advantage); the weight-GAINING diabetes drugs are sulfonylureas, TZDs, and insulin.

2. DPP-4 inhibitor

WHAT YOU SEE

DPP-4 with red X enzyme scissors

WHAT IT ENCODES

DPP-4 inhibitors (the '-gliptins': sitagliptin, saxagliptin, linagliptin, alogliptin) block degradation of endogenous incretins (GLP-1/GIP), raising their levels. They are oral, weight-NEUTRAL, modest A1c-lowering (~0.5-0.8%), and carry low hypoglycemia risk. Saxagliptin/alogliptin are associated with increased heart-failure hospitalization, so prefer other gliptins in HF patients.

BOARD TRAP

Wrong answer: 'add a DPP-4 inhibitor to a patient already on a GLP-1 receptor agonist.' Discriminator — both act on the incretin axis; combining a DPP-4 inhibitor with a GLP-1 RA is redundant (no added benefit) and not recommended. Choose one incretin-based agent.

3. GLP-1 angioedema

WHAT YOU SEE

A small swollen-faced portrait sign tagged ANGIOEDEMA on the gut wall — the puffy face is the rare hypersensitivity reaction that means STOP the incretin.

WHAT IT ENCODES

Incretin therapies can rarely cause hypersensitivity reactions including angioedema and anaphylaxis. Discontinue the drug if a serious hypersensitivity reaction occurs and treat supportively; do not rechallenge.

BOARD TRAP

Wrong answer: 'angioedema after starting therapy is just an injection-site reaction — continue the drug.' Discriminator — angioedema is a systemic hypersensitivity reaction that warrants stopping the agent, distinct from a benign local injection-site reaction.

4. GLP-1 GI / pancreatitis & MTC

WHAT YOU SEE

The cluster of harm-tags around the GLP-1 jar in the gut tube — the swollen ANGIOEDEMA face plus the pancreas figure slumped sick on the dock — flagging the class GI/pancreatitis and MTC/MEN2 warnings.

WHAT IT ENCODES

The dominant GLP-1 RA side effects are GI (nausea, vomiting, diarrhea) from slowed gastric emptying — usually transient with slow titration. Class warnings: pancreatitis (stop for severe persistent abdominal pain) and a boxed contraindication with personal/family history of medullary thyroid carcinoma (MTC) or MEN2. Also avoid in established gastroparesis.

BOARD TRAP

Wrong answer: 'prescribe a GLP-1 agonist to an obese diabetic with a family history of medullary thyroid carcinoma.' Discriminator — personal or family MTC/MEN2 history is an absolute contraindication to GLP-1 RAs and tirzepatide (not to DPP-4 inhibitors); pick a different class.

5. Joints (DPP-4)

WHAT YOU SEE

A red-flagged JOINTS placard pinned next to the DPP-4 scissors — the aching joint marks the severe disabling arthralgia that gliptins can cause.

WHAT IT ENCODES

DPP-4 inhibitors can cause severe, sometimes disabling arthralgias (a recognized class adverse effect) and, rarely, acute pancreatitis. If a patient develops severe joint pain after starting a gliptin, consider the drug as the cause and discontinue.

BOARD TRAP

Wrong answer: 'new severe polyarthralgia on a gliptin must be new inflammatory arthritis — work it up and continue the drug.' Discriminator — DPP-4 inhibitors are a recognized cause of severe arthralgia that resolves on discontinuation; stop the gliptin before an extensive rheumatologic workup.

6. Linagliptin safe in renal disease

WHAT YOU SEE

LINAGLIPTIN safe in renal disease tag

WHAT IT ENCODES

Linagliptin is primarily excreted via the bile/enterohepatic route, so it requires NO renal dose adjustment and is usable even in advanced CKD/ESRD. The other gliptins (sitagliptin, saxagliptin, alogliptin) are renally cleared and need dose reduction as eGFR falls.

BOARD TRAP

Wrong answer: 'use full-dose sitagliptin in a patient with eGFR 20 because all gliptins are renally safe.' Discriminator — only linagliptin needs no renal adjustment; the renally cleared gliptins require dose reduction in CKD to avoid accumulation.

7. Alpha-glucosidase inhibitor soldier

WHAT YOU SEE

Soldier chaining CARBS, alpha symbol, POST-PRANDIAL

WHAT IT ENCODES

Alpha-glucosidase inhibitors (acarbose, miglitol) block intestinal brush-border enzymes, slowing carbohydrate digestion and blunting POST-PRANDIAL glucose spikes. They are weight-neutral with modest efficacy; main side effects are flatulence, bloating, and diarrhea. Key point: treat any hypoglycemia (when combined with insulin/sulfonylurea) with oral GLUCOSE, not sucrose, because sucrose breakdown is blocked.

BOARD TRAP

Wrong answer: 'treat hypoglycemia in a patient on acarbose with table sugar (sucrose).' Discriminator — acarbose blocks sucrose hydrolysis, so sucrose won't raise glucose; use pure glucose (dextrose) tablets/gel instead.

8. SGLT2 inhibitor plumber

WHAT YOU SEE

Plumber draining glucose with wrench, SGLT-2, 90 percent reabsorption

WHAT IT ENCODES

SGLT2 inhibitors (the '-flozins': empagliflozin, dapagliflozin, canagliflozin, ertugliflozin) block the proximal-tubule SGLT2 cotransporter that reabsorbs ~90% of filtered glucose, causing glucosuria and natriuresis. This lowers glucose insulin-independently (low hypoglycemia risk), produces mild weight loss and BP reduction, and confers cardiorenal protection.

BOARD TRAP

Wrong answer: 'an SGLT2 inhibitor will cause hypoglycemia like a sulfonylurea.' Discriminator — SGLT2 inhibitors act independently of insulin secretion, so monotherapy rarely causes hypoglycemia; the secretagogues (sulfonylureas, meglitinides) and insulin are the hypoglycemia culprits.

9. SGLT2 heart failure / CKD benefit

WHAT YOU SEE

HF and CKD ribbons on SGLT2 plumber

WHAT IT ENCODES

SGLT2 inhibitors reduce heart-failure hospitalizations (benefit in HF_{rEF} AND HF_{pEF}, even without diabetes) and slow CKD progression / reduce albuminuria and renal endpoints. They are the preferred add-on (alongside or after metformin) when HF or CKD coexists with T2DM. Cardiorenal benefit persists at lower eGFR even as glucose-lowering wanes.

BOARD TRAP

Wrong answer: 'pick a DPP-4 inhibitor (or sulfonylurea) for glycemic control in a diabetic with heart failure and CKD.' Discriminator — when HF or CKD is present, choose the agent with proven cardiorenal benefit (SGLT2 inhibitor); DPP-4 inhibitors lack renal/HF outcome benefit (and saxagliptin can worsen HF).

10. Euglycemic DKA

WHAT YOU SEE

EU-DKA warning triangle, GLU 180

WHAT IT ENCODES

SGLT2 inhibitors can precipitate euglycemic DKA — a high-anion-gap ketoacidosis with glucose that is only mildly elevated or near-normal (often <200-250 mg/dL). Triggers include surgery, fasting, acute illness, low-carb/ketogenic diets, and alcohol. Check ketones/anion gap in any unwell SGLT2-treated patient regardless of glucose.

BOARD TRAP

Wrong answer: 'this acidotic SGLT2-inhibitor patient can't be in DKA because the glucose is only 180.' Discriminator — SGLT2-associated DKA is euglycemic; a near-normal glucose does NOT exclude DKA. Measure ketones and the anion gap.

11. Sick-day / hold rule

WHAT YOU SEE

SICK DAY HOLD tag

WHAT IT ENCODES

To prevent euglycemic DKA, HOLD SGLT2 inhibitors during acute illness, prolonged fasting, and around surgery — guidance is to stop ~3-4 days before elective surgery (ertugliflozin ~4 days; longer-acting agents per protocol) and resume only when eating and clinically stable. This 'sick-day rule' also applies to severe dehydration or volume depletion.

BOARD TRAP

Wrong answer: 'continue the SGLT2 inhibitor through the perioperative fasting period to keep glucose controlled.' Discriminator — perioperative fasting plus SGLT2 inhibition is a classic euglycemic-DKA trigger; the drug must be held before surgery, not continued.

12. Fournier gangrene / genital infection

WHAT YOU SEE

YEAST + FOURNIER skull tags at base of pipe; repeated on right panel

WHAT IT ENCODES

Glucosuria from SGLT2 inhibitors fosters genital mycotic infections (vulvovaginal candidiasis, balanitis) and urinary tract infections, plus the rare but life-threatening Fournier gangrene (necrotizing fasciitis of the perineum). Counsel on genital hygiene; severe perineal pain/erythema/swelling with systemic illness is a surgical emergency.

BOARD TRAP

Wrong answer: 'severe perineal pain and erythema in an SGLT2-treated patient is just a routine yeast infection — give topical antifungal.' Discriminator — rapidly progressive perineal pain with systemic toxicity suggests Fournier gangrene, a surgical emergency requiring urgent debridement, not outpatient antifungal.

13. SGLT2 not in T1DM

WHAT YOU SEE

NO T1DM tag on right panel

WHAT IT ENCODES

SGLT2 inhibitors are not recommended in type 1 diabetes because the insulinopenic state markedly raises the risk of (euglycemic) DKA. They are agents for T2DM (and have HF/CKD indications); in T1DM, basal-bolus insulin remains the foundation.

BOARD TRAP

Wrong answer: 'add an SGLT2 inhibitor to a T1DM patient to reduce insulin needs and aid weight loss.' Discriminator — in T1DM the DKA risk from SGLT2 inhibition is unacceptably high; this is the wrong class for T1DM.